Zeros of Polynomial Functions

Name _____

Algebra 2 Lesson 3-5 (fast track) Date _____ Period _____

Learning goal: I can find the zeros of a polynomial function by factoring and use them to sketch its graph.

Language goal: I can explain what a graph does at each zero using the words zero, factor, multiplicity, crosses, and turning point.

On the quiz: (1) Factor a cubic, list all its zeros, and sketch its graph through them (Example 1). (2) Solve a polynomial equation (full Savvas quiz only). (3) Solve a polynomial inequality (full Savvas quiz only). (4) Read the zeros and their multiplicities off a graph (Example 2). (5) Find the zeros of a cubic with a double zero and pick the right description of its graph: end behavior, above or below the x-axis between zeros, where it touches the axis (Examples 1 and 2, then the Practice).

On the Topic 3 test: Question 11: does $f(x) = (x-5)(x-1)(x+4)$ cross or touch the x-axis at each zero (Example 2)? Question 12: the zeros of $f(x) = x(x-3)(x+1)$ (Example 1).

Example 1: Use Zeros to Graph a Polynomial Function

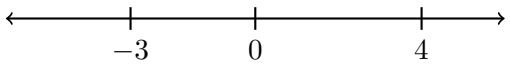
What are the zeros of $f(x) = x(x-4)(x+3)$? Graph the function.

1. Zeros. A **zero** is an x-value where $f(x) = 0$. By the Zero-Product Property, a product is 0 only when a **factor** is 0:

$x = 0$ $\quad x - 4 = 0 \Rightarrow x = 4$ $\quad x + 3 = 0 \Rightarrow x = -3$

The zeros are -3, 0 and 4.

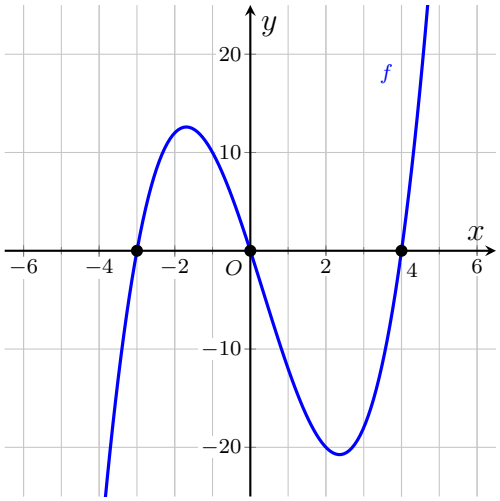
2. Intervals. The zeros cut the number line into four intervals.



3. Signs. Pick the sign of each factor on each interval; the product's sign says where the graph is.

Interval	x	$x-4$	$x+3$	**Product**	**Graph**
$x < -3$	−	−	−	−	below
$-3 < x < 0$	−	−	+	+	above
$0 < x < 4$	+	−	+	−	below
$x > 4$	+	+	+	+	above

4. Sketch. Draw one smooth curve through each zero: below the x-axis where the product is negative, above it where the product is positive.



Check: $f(-2) = (-2)(-6)(1) = 12$ and $f(2) = (2)(-2)(5) = -20$, so the curve really is above, then below.

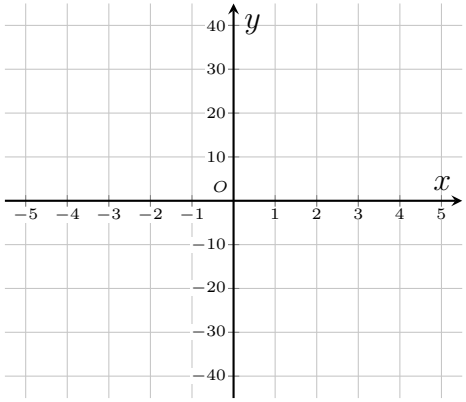
Say it: The **zeros** of f are _____ because each **factor** _____. On $-3 < x < 0$ the graph is _____ the x-axis because _____.

Try It 1: Factor each function. Then use the zeros to sketch its graph.

a. $f(x) = 4x^3 + 4x^2 - 24x$

Factor: $f(x) =$ _____

Zeros: _____

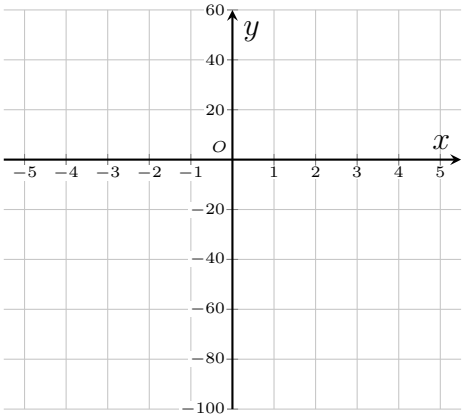


b. $g(x) = x^4 - 81$

Hint: $x^4 - 81 = (x^2)^2 - 9^2$; the factor $x^2 + 9$ is never 0 for a real x.

Factor: $g(x) =$ _____

Zeros: _____



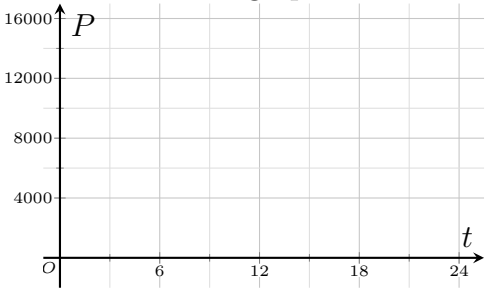
Sign table for part a (one row per interval):

Interval	factor	factor	factor	Product	Graph

Extension: Practice 30 (DOK 3), when you finish

Performance Task Venetta opened several deli sandwich franchises in 2000. The profit P (in hundreds of dollars) of the franchises in t years (since the franchises opened) can be modeled by the function $P(t) = t^3 + t^2 - 6t$.

Part A Sketch a graph of the function.



Part B Based on the model, during what years did Venetta not make a profit?

_____

_____

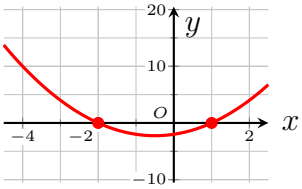
Part C If the model is appropriate, predict the amount of profit Venetta will receive from her franchises in 2020.

_____

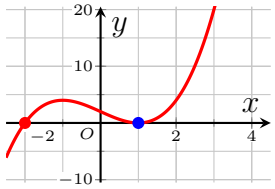
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Example 2: Understand How a Multiple Zero Can Affect a Graph

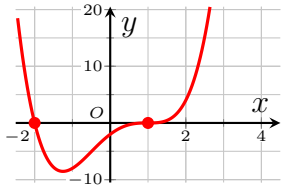
How does a multiple zero affect the graph of a polynomial function? The **multiplicity** of a zero is the number of times its related factor appears in the factored form of the polynomial. Watch each graph as it reaches $x = 1$.



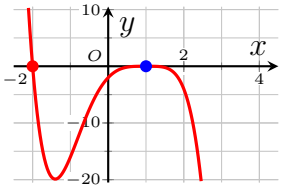
$f(x) = (x-1)(x+2)$



$f(x) = (x-1)^2(x+2)$



$f(x) = (x-1)^3(x+2)$



$f(x) = -(x-1)^4(x+2)$

Function	$(x-1)(x+2)$	$(x-1)^2(x+2)$	$(x-1)^3(x+2)$	$-(x-1)^4(x+2)$
multiplicity of 1	1 (odd)	2 (even)	3 (odd)	4 (even)
graph at $x = 1$	crosses	turning point	crosses (flattens)	turning point
multiplicity of -2	1 (odd)	1 (odd)	1 (odd)	1 (odd)
graph at $x = -2$	crosses	crosses	crosses	crosses

When the **multiplicity** of a zero is **odd**, the graph **crosses** the x-axis. When the multiplicity is **even**, the graph has a **turning point** at the x-axis (it touches and turns back). Close to a zero, the graph looks like a polynomial whose degree is the multiplicity: multiplicity 3 crosses with a flat "x^3" bend.

Say it: In $f(x) = (x-1)^2(x+2)$ the **factor** $(x-1)$ appears _____ times, so the **zero** 1 has **multiplicity** _____, which is _____. The graph has a _____ at $x = 1$, and it _____ the x-axis at $x = -2$.

Try It 2: Behavior at each zero

Describe the behavior of the graph of the function at each of its zeros.

a. $f(x) = x(x+4)(x-1)^4$

zero	**factor**	**mult.**	**behavior**

b. $f(x) = (x^2+9)(x-1)^5(x+2)^2$
Hint: x^2+9 is never 0 for a real x, so it adds no zero to the graph.

zero	**factor**	**mult.**	**behavior**

Practice: From standard form to the shape of the graph

$f(x) = -x^3 + x^2 + 8x - 12$

a. Show that 2 is a zero: $f(2) =$ _____

b. Divide by $x - 2$ (synthetic division), then factor completely.

2	-1	1	8	-12
		____	____	____
	____	____	____	____

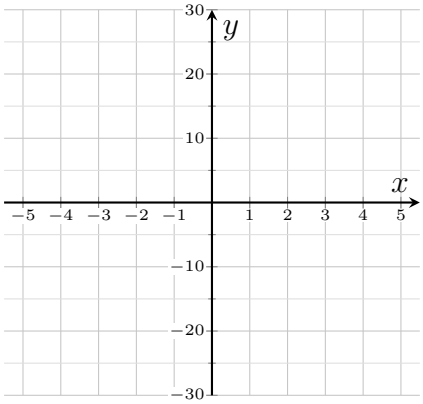
$f(x) = (x - 2)\Big($_____$\Big) =$ _____

c. Zeros and multiplicities: _____

d. Finish each statement.

- As $x \to -\infty$, $f \to$ _____; as $x \to \infty$, $f \to$ _____.
- When $-3 < x < 2$, f ____ 0 (test: $f(0) =$ _____).
- At $x = 2$: multiplicity ___, graph _____.
- At $x = -3$: multiplicity ___, graph _____.

e. Sketch f through its zeros.



IXL: Practice for the quiz

Go to IXL, Algebra 2, and type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
PVM ★	Find the zeros of factored polynomials	Quiz 1A, 4, 5 (Examples 1, 2)
8CQ ★	Match polynomial functions and graphs using zeros	Quiz 1B, 4 (Examples 1, 2)
NCY ★	Factor a polynomial given a root	Quiz 5 (the Practice)
UB5	Factor quadratics	Warm-up for quiz 1A
XMK	Match polynomial functions and graphs using end behavior	Warm-up for quiz 5

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Before the quiz: Finish these sentences

If $x = a$ is a **zero** of f, then _____ is a **factor**, and the graph meets the x-axis at _____.

A zero with odd **multiplicity**: the graph _____ the x-axis. Even **multiplicity**: the graph has a _____ there.

Between two neighboring zeros the graph stays _____, so I test one x-value.